Training a Playlist Curator Based on Track Features

UE24CS352A - Machine Learning Mini-Project | Team: Mohammed Adaan Hamad & Lakshya Jeet Singh | Section: E

Problem statement

Given the audio features of a track, recommend the playlist genres that best fit it. A track can appear in multiple playlists, so we model genre assignment as a multi-label problem. The project adapts the supplied CS229 playlist-curation idea to the provided Spotify Songs dataset.

Dataset

The supplied archive contains 32,833 playlist-track rows and 23 columns. After grouping repeated rows by track ID, the project uses 28,356 unique tracks, 12 audio features, and six playlist genres: edm, latin, pop, r&b;, rap, rock. We found 1,687 tracks with more than one genre label. The features are danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration. We exclude track popularity and playlist metadata from the model inputs.

Approach

We average repeated audio feature values by track ID and retain every associated playlist genre. We split unique-track rows into 22,684 development and 5,672 test tracks (80/20, seed 42). The test set remains untouched during model selection. We compare one-versus-rest logistic regression, one-versus-rest RBF SVM, and a multi-output random forest using 5-fold cross-validation on the development set. Median imputation and scaling are fit inside each fold's training pipeline. We compare micro F1, macro F1, exact subset accuracy, top-1 hit rate, and recall at three; mean macro F1 selects the model.

Implementation and measured results

The Python pipeline validates and reads the CSV, aggregates playlist rows at track level, builds multi-hot genre labels, trains and compares models, and saves metrics, held-out predictions, and the selected model. The Streamlit demo accepts a CSV with the same 12 features and ranks the three most likely playlist genres.

Model	Micro F1	Macro F1	Top-1 hit	Recall@3
Logistic Regression	0.463	0.473	0.485	0.826
RBF SVM	0.525	0.533	0.548	0.865
Random Forest	0.572	0.563	0.589	0.878

Selected model: **Random Forest** by mean cross-validation macro F1. On the untouched test set, it achieved macro F1 0.563 and micro F1 0.576. Exact subset accuracy requires all genre labels for a track to match.

Conclusion

The selected model (Random Forest) provides the strongest macro-F1 result on this fixed track-level test split. The project demonstrates how acoustic features can rank broad playlist genres, including multiple plausible genres for a track. The task does not measure personal taste or reproduce a specific listener's playlist decisions. Evaluation on listener-curated playlists and repeated splits would provide stronger evidence.

Limitations and data note

The archive README says the dataset was obtained from Spotify via the spotifyr package and does not specify a redistribution license. Spotify's current Developer Policy restricts using Spotify content to train ML models. We did not call the Spotify API. Confirm with the instructor that use of this supplied archival dataset is permitted for the course, and do not redistribute it without permission.

References

Awadelkarim and Coelho (2018), Training a Playlist Curator Based on User Taste, CS229 report and poster. Defferrard et al. (2017), FMA: A Dataset For Music Analysis, ISMIR, https://arxiv.org/abs/1612.01840 (alternative open music dataset). Spotify for Developers, Spotify Developer Policy, effective 15 May 2025, https://developer.spotify.com/policy.